

Chapter 10

From Mechatronic Systems to Cyber-Physical Systems: Demands for a New Design Methodology?

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10.1 From Mechatronic Systems to Cyber-Physical Systems

Mechatronics may be defined as an interdisciplinary field of engineering science which aims to mutually integrate and interconnect mechanical engineering, electrical engineering/electronics and computer science (also often called information technology) such that the interactions constitute the basis for the design of successful products [1–3].

In a mechatronic design process the conceptual design phase is especially important. Here, the functional interactions between discipline-specific sub-systems are determined, and therefore have to be carefully considered. This implies that during the phases of conceptual and preliminary design, the designer should be able to quickly and accurately evaluate those system properties that result from design changes in the mechanical components as well as in other components.

The successful development of complex mechatronic systems is only possible by close cooperation between specialists in the different disciplines involved. Thus the design activities take place in a multidisciplinary environment, which often involves engineers and other experts with different backgrounds [4]. In order to enhance the performance of new products, the positive interaction between different fields of mechatronics is increasingly being used. This may result in an increased complexity of the product because the mere combination

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of discipline-oriented partial solutions will usually not provide the optimal result attainable with an integrated system [5].

Mechatronic design emphasizes the integration between engineers skilled in specific domains such as mechanics, electronics and software. The interactions between product developers from these different disciplines are, however, often hindered by insufficient understanding between the disciplines, and by missing common platforms for modelling of complex systems [6]. As many sub-systems are sourced from external suppliers, there is a need for both horizontal integration within organizations and for vertical integration between the sub-system suppliers and the suppliers of the full systems. Lee [7] defines Cyber-Physical Systems (CPS) as the:

.... integration of computation and physical processes. Embedded computers and networks monitor and control the physical processes, usually with feedback loops where physical processes affect computations and vice versa.

Such systems can be found in areas such as aerospace, automotive, energy, healthcare, manufacturing, entertainment and consumer appliances.

For easy validation of requirements during the overall product development process methods are needed for the decomposition of high-level system requirements into criteria for design decisions. This will be an important aspect in the future, where new system architectures such as those of CPS are introduced [8, 9].

System modelling and evaluation are also important topics of CPS, for which improved tools and knowledge are always required by the engineering profession. In many cases, very accurate system modelling is not a reasonable approach to describing a complex system as the uncertainties and costs of even relatively detailed modelling may be so high that the drawbacks compared to simpler modelling become overwhelming, so there is increasing trend towards system-level models which allow a multidisciplinary engineering approach to be supported.

CPS system-level models need specific methods, languages and tools to support multi-view modelling in order to facilitate an interdisciplinary approach. More generally, this objective can be realized through multiagent modelling, based on an engineering cloud structure. This also results in the usage of tools supported by Model-Based Systems Engineering (MBSE) [10].

CPSs are often dominated by one engineering discipline. System-level models have to promote an equal treatment of all engineering disciplines involved during product development and project execution.

Figure 10.1 presents a structure model describing the relationships between mechatronic systems and CPS (shades of grey). These core elements are all comprised of a hardware element and a software element. The interactions between these can occur in the physical domain (e.g. clash between two robots detected, thanks to their sensors), or in the cyber part (e.g. dialogue between these robots supported by network protocols). The cyber part is then considered as the integration network. All these core elements are themselves made up of several modules represented by the small white boxes. The mechatronic systems and the CPS located on the border are then parts of the Systems of Systems.

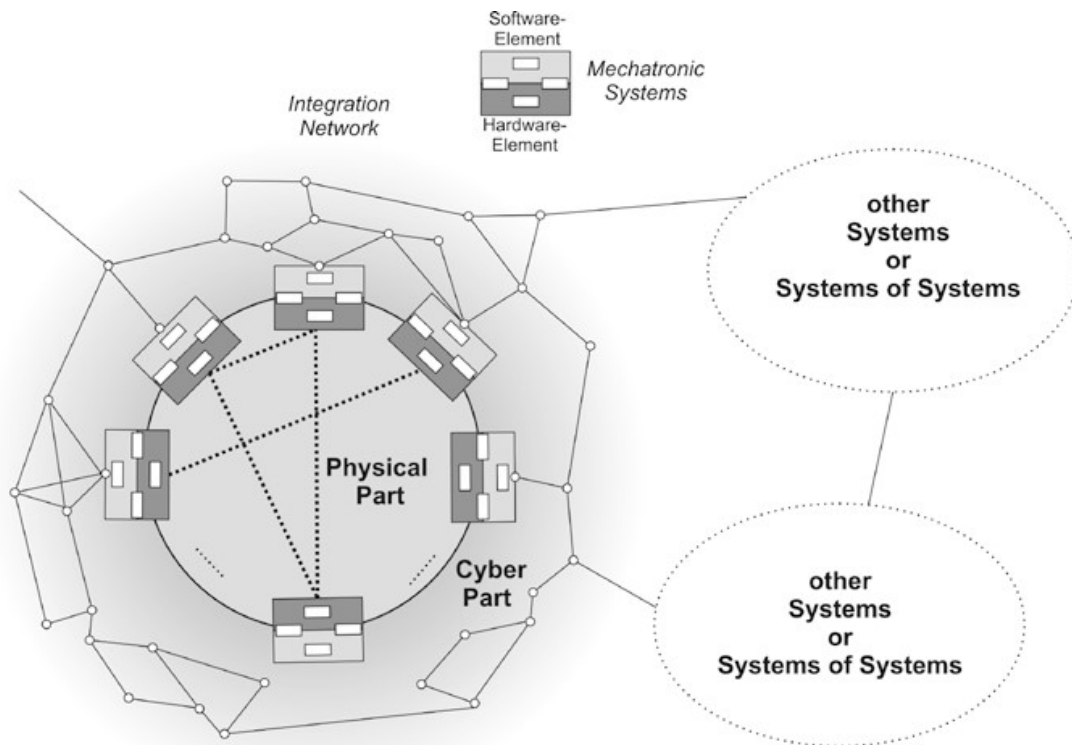


Fig. 10.1 Mechatronics and cyber-physical systems as a systems of systems [6]

10.2 System Design Methods

In this section the issue of system design methods is presented considering the hierarchy of systems and the used models.

10.2.1 Hierarchical Modelling

One of the major challenges in developing mechatronic products is the increasing function density and complexity of such products as well as of the corresponding mechatronic design processes.

The defining feature of mechatronic products, also known as systems of systems, integrated systems, or mixed systems, is that they merge solutions from disparate disciplines. As a consequence, a mechatronic design process must integrate multiple disciplines. There is a significant lack of such integration in traditional design processes which in general have emerged more or less from a single discipline and have incorporate other disciplines later.

The mechatronic design approaches (e.g. the V-Model of VDI 2206) focus on exactly this integration aspect of the synthesis, analysis and evaluation steps in modern mechatronic design. However, there are still a lot of open questions.

Hierarchical modelling concepts, i.e. the use of models with different granularity for different levels of abstraction, are a promising approach to model and master complex systems. The chosen approach is based on a modular structure of models (modular model architecture, model base, hierarchical structure of models) that allows for the configuration of system models from a library of sub-models and interface models. A sub-model can be a model of a single mechanical component or of a complex system, e.g. a model of an integrated mechatronic system including an embedded control system. An interface model represents the (physical) interaction between two sub-models, between sub-models and the system model or between the system model and (the model of) the environment of the system under consideration. This results in the advancement of reduced order modelling as a key for coping with models of complex systems, the improvement of the system view by system modelling and a significant advancement of the mechatronic design process itself by more systematic approaches with particular focus on the early phases of design (conceptual design, preliminary design).

The structure of the model of a CPS aims to store the product information from the entire product life cycle. This requires a hierarchical structure, which can be used in the early phases, which does not change during the product development process. Normally, product information is structured according to either the geometry or the assembly structure of the product. This results in problems in the early phases of the product life cycle as the geometry or assembly structure results from the development process and neither exists in the early phases nor is stable during the development process. The object which is most stable during product development is the set of resulting properties which define the product. As previously stated, requirements can change over time, but to be precise, changed requirements result in the definition and development of a new product. Trying to find a product information structure which is suitable for every set of requirements ultimately results in a structure which is identical for every product development process [11].

On the other hand, we have network of interactions and properties influenced by the different system elements. Basically, one type of property, a definable property, can be any property the designer defines directly (e.g. materials, manufacturing parameters, geometry). The totality of all definable properties then defines the complete product with all its properties and its behaviour.

The resulting properties are used to structure the generally high number of definable properties. This is done by assigning each definable property to the resulting properties influenced by this definable property. As previously mentioned, it is possible that a definable property influences more than one resulting property. For example, the definable property “*material*” influences the resulting properties “*maximum weight*” and “*maximum stress*”). A second level of structuring is achieved by assigning the definable properties to the different views. Each definable property can appear in a single view or in multiple views. For instance, the definable property “*material*” appears in the views “*producibility*” and “*costs*”. In order to classify the definable properties in the matrix, a meta-information can be assigned to each of them. In most cases more than one model can be used for

the description of a specific resulting property, whereas the less complex subsidiary models were used in the early phases of the product development process, when there is only little information about the product. During the product development process more and more complex subsidiary models can be used, to describe the interrelation between definable properties and resulting properties.

The knowledge about the interrelation between definable properties and resulting properties, gathered together from the subsidiary models can be used to set the values of the definable properties, such as geometric dimensions, material type, properties of the production processes. For example, the product designer uses a FEM model to determine stress.

10.2.2 System Model

During all phases of the design process there is a need to build models which may be seen as simplified representations of an original. In different phases these design models have different goals. During the conceptual design phase, physical principles, functions, structures, etc., have to be evaluated by executing models. In the context of mechatronic system design processes, the phenomena under consideration usually are of a physical or chemical nature. The models consist of a set of parameters as well as a set of logical and quantitative relationships between these models [4, 12].

The number of parameters increases from the conceptual design stage to preliminary and detailed design. According to the increasing degree of detailing during the design process, the granularity of the describing models becomes finer and finer, leading to a hierarchy of models as well as their describing parameters. For modelling and evaluation of solutions during all phases of the design process, we postulate models with different degrees of detailing (granularity of models) in correspondence with their describing parameters. The correspondence between models and parameters implies that the meaning of a parameter is well enough defined via its related model. Hierarchical models are very important tools for complex activities such as engineering design. Especially during the conceptual design phase where there is a high demand for models to describe the design concept with respect to the given requirements.

In addition to the hierarchical differentiation, the extensive functionality and complex structure of mechatronic systems has the result that it is generally not enough to optimize on a single criterion and a multi-objective optimization is often needed. Optimization separately within each domain will not result in the optimum system design. Therefore all the domains of, for instance, an automotive sub-system have to be treated concurrently, at least in the beginning of the design process. That way it will be possible to translate aspects from say the control or electrical design to the mechanical design at an early stage of the design process. This approach makes it possible to find a promising concept for the entire sub-system and not only for a specific domain within the sub-system.

In order to master the mechatronic design approach and to benefit from it as much as possible, a hierarchical design process is proposed in [11], in which the discipline-specific design tasks need not be integrated as a whole on the mechatronic level of the design task. Consequently, the system models should cover the different views on a system as well as the different degrees of detailing which lead from a hierarchy of models to a hierarchy of design parameters. Their description by significant quantities is used for analysing different mechatronic design concepts. The general requirements can be summarized as:

- Description of product information from all phases of the product life cycle taking into account the lifespan of a product as a necessity when storing the product relevant information within a single structure.
- Coupled analysis of different physical product properties as caused by the increasing integration of different applications and functionality in technical systems, the demand arises to find a uniform description of the various product properties, to define, describe, dimension, and calculate the product.
- Taking into account the view on an application domain as different development phases and mechatronic disciplines require different views on the object to be analysed. A general description of the requirement entails that model parameters should be provided in various combinations and different granularity for a further (more detailed) processing.

A major characteristic of mechatronic systems is that their properties are to a considerable extent defined by software elements. This initially leads to a shift from the physical function of the basic system to a realization through electronics or information technology. As a consequence, with some adaptations, intelligent systems of the future could include learning and decision-making abilities. Such systems may include, for example self-optimizing processes to adapt to changing conditions.

Crucial to the success of such a product is the behaviour of the overall system, as customer requirements and desires almost always relate to the whole system and not to partial sub-systems, components or even individual components. To assess the characteristics of the system it is appropriate to use models. For modelling and description of a mechatronic system, it is necessary to decompose the system into a selection of suitable subdomains, describing the boundaries of the considered mechatronic system to its system environment, enabling the flow of matter, energy and information.

The functioning of mechatronic systems is distinguished from other systems and hence requires the clear definition of interfaces and areas of responsibility. It must therefore be clarified in what way interactions with the system environment (e.g. chemical, energy, information) have to take place. In the ideal case, the whole system has the form of a cross-domain model. Unfortunately, the different disciplines use different modelling approaches and model descriptions. Moreover, within the disciplines, information and data at high detail level are only partially needed across the other disciplines. It is necessary to collect data for the overall behaviour for understanding of the overall relevance. The objective is to

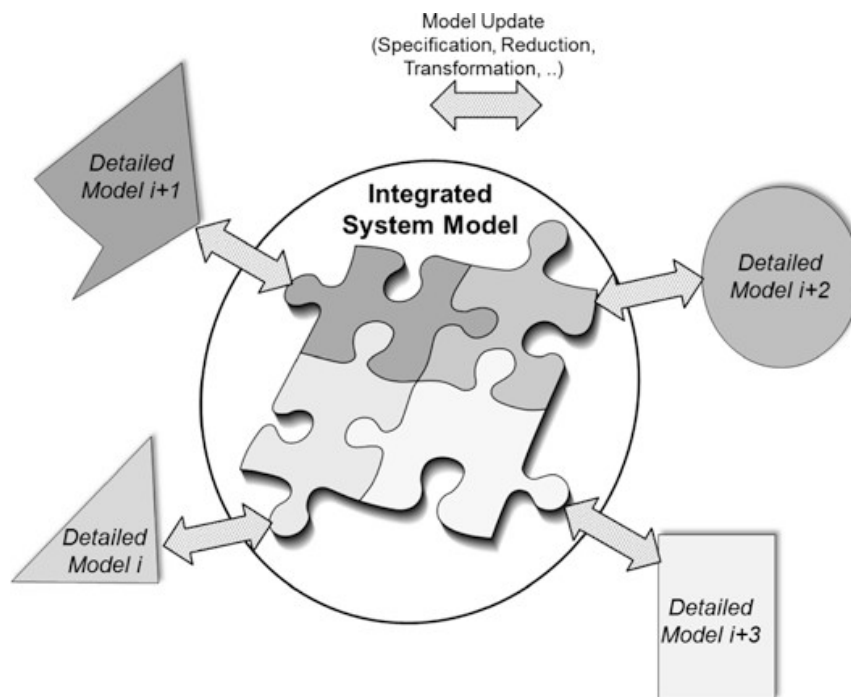


Fig. 10.2 System-level models [21]

create a system model with the information depicting the various domains, with focus on the importance for other domains. The challenge is that the knowledge of the entire system does not equal the sum of knowledge from the corresponding domains. The domain knowledge must therefore be generalized (abstracted) and integrated (see Fig. 10.2).

System-level models should at least facilitate management of existing data and visualization of both the relationships inside a system (between its sub-systems) and between a system and its environment. Additionally, they should provide the possibility to execute several simulations of load cases (test cases), thus allowing specific system properties to be evaluated. The simulations at the system level differ from those at the discipline level. Since simulations at the discipline level are usually conducted by highly skilled and specialized engineers who use particular, discipline-specific software tools, the simulations at the discipline level normally cannot be replaced by simulations at the system level. Therefore, methods are necessary:

- For modelling and simulation with special emphasis on the system view of the design object, i.e. the object under design.
- For conceptual and preliminary design relying on concept models at an appropriate system level representing the essential information including a significant system view of the design object.
- For the definition of modular structure of models (modular model architecture, model base, hierarchical structure of models) that allows for the configuration of system models from a library of sub-models and interface models.

- For the decision as to which information (e.g. which properties and which of their quantifying parameters) should be included in the system model and which in the sub-models.

10.3 Robust Design of Mechatronic Products and Systems

In this section the issue of robust design of mechatronic products is looked at, and several challenges are raised with respect to the differences in perspectives of the mechanical and electrical disciplines when it comes to reliability and robustness.

10.3.1 Motivation for Robust Design

Robustness relates to the ability of a product to perform despite unwanted variation to parameters and noise factors. Robust design is therefore an incredibly important methodology in mechanical design as it has the potential to allow manufacturing and assembly to have much looser tolerances, driving down the cost of production and increasing the quality of the product.

The benefits of having a robust design are not just limited to the relaxing of tolerances. In the news headlines and magazine reports, it is common to see stories of product recalls, high customer complaint rates, reduced product reliability, unscheduled maintenance and repair and even major or catastrophic product failure, all being damaging symptoms of a lack of product robustness.

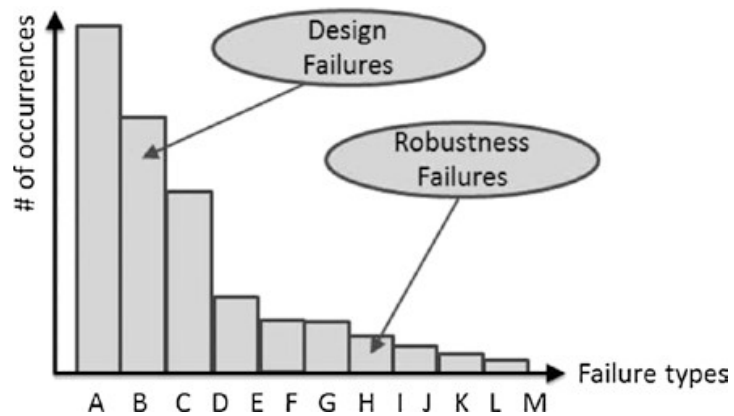
However, this is the tip of the Iceberg! The reliability and quality issues that occur within a company due to a lack of robustness are large and costly. They include delayed product launch dates, misplaced RD resources, increased quality control and inspection and reduced innovation height.

While each discipline has its own approach to handling robustness, it is now very common for the effects of variation to run from one discipline through to the next, requiring robust design and optimization across disciplines, most typically from the mechanical to the electrical domains.

10.3.2 Robustness as Hidden Reliability

It is clear from interacting with the reliability engineering community, that the electrical engineers have a different tool sets and approaches to those adopted by mechanical engineers with regards to product reliability. However, there are some clearly overlapping methods.

Fig. 10.3 Occurrence of failure modes [13]



For instance, Highly Accelerated Life Testing (HALT) is a methodology that can test the reliability of both the mechanical and electrical aspects of a design both independently and integrated into a system. HALT is used to test a single product (perhaps of nominal dimension) often in its intended use conditions, but with a highly accelerated use cycle. If we look at the nature of product failures in industry, HALT will only give part of the story. Figure 10.3 [13] is an illustrative example of the occurrence of the different failure modes. The failure modes with high occurrence (labelled design failures) are easily detectable through product testing such as HALT. However, the long tail of failure modes may occur perhaps only one in 100 or one in 1000 and may not be detectable during testing. These types of failures are termed robustness failures.

In addition to the above, it is clear that electronics reliability testing is dominated by Weibull analysis and mean time to failure. This is dominated by the more binary nature of electronics—it is working or it is not. Weibull would help to make a probabilistic failure analysis of components and therefore with a fault tree analysis (or similar) to calculate the probability of system failure at a given time.

In mechanical design, approaches such as kinematic design, design clarity [14], location scheme design [15], axiomatic design [16] are typical approaches to dampen issues when the product is not nominal (variation is present). The focus here is not to predict product life but to predict and decrease the change in performance due to variation. In an additional publication [17], 15 strategies for variation reduction are laid out. However, only two of these strategies (Quality Control and Shielding) are possible in the electronics industry.

10.3.3 Electrical Mechanical Interfaces

All electrical components have a mechanical interface to them. Mounting and fitting of PCBs is notorious for throwing up late stage issues and failures especially during the high volume ramp-up. There are two common errors here. First it seems to be a common mistake to overconstrain the PCB within the housing. An example

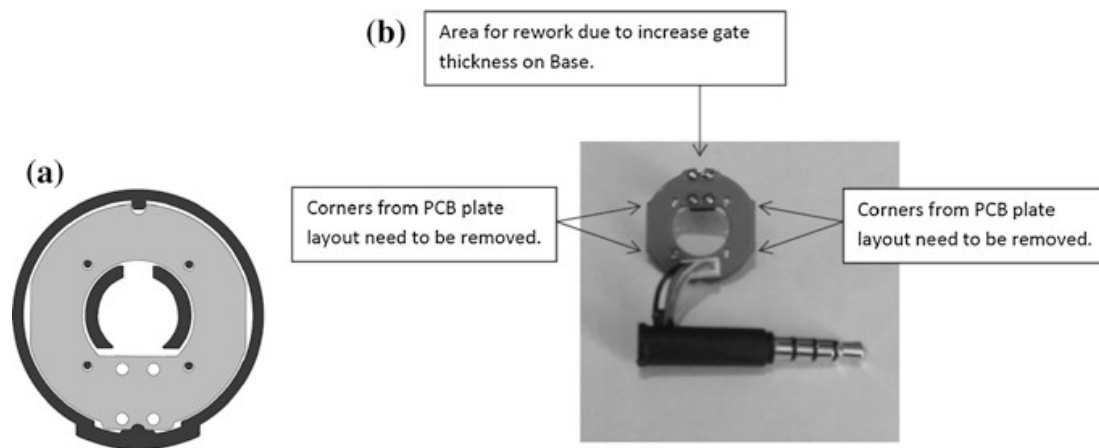


Fig. 10.4 Effect of overconstraint of PCB and housing. **a** Original design. **b** Redesign

of this can be seen in Fig. 10.4a where it is clear that there are too many surfaces responsible for the positioning of the PCB within the housing (the outer edge, the four pins, the inner edge, the notch at the top and the bottom). As a consequence the PCB had to be reworked in various areas before the assembly process—the areas of rework can be seen in Fig. 10.4b.

In the case above where a PCB is unable to fit correctly in its housing, the worst case will be time and money spent on rework before assembly. In some cases it may also lead to a potential distortion of the PCB during mounting, causing it some damage. When it comes to mounting of sensors, the positioning of the sensor can have a huge consequence on the performance of the product, this is a case of cross-domain robustness, dealt with in the next section.

10.3.4 Cross-Domain Robustness

With reference to mechatronics, cross-domain robustness arises where a variation in the mechanical domain leads to performance change in the electrical domain, or variation in the electrical domain affects the mechanical performance. There are many examples of this in various signal processing applications. One typical example can be seen in Fig. 10.5 where a series of teeth are arranged to rotate through a detector. The counting of the teeth by the detector determines both the speed of rotation and the position of the rotating part.

Figure 10.5b shows how the teeth are returned from manufacturing where the flash from the injection moulding introduces unwanted dimensional variation on the teeth, which means that the signal generated was less precise and may even reach a failure mode where a tooth is not recognized, producing a step change in the velocity reading.

The signal produced from the detector and the tooth arrangement in Fig. 10.5a is shown in Fig. 10.6. The signal processing software will then process the

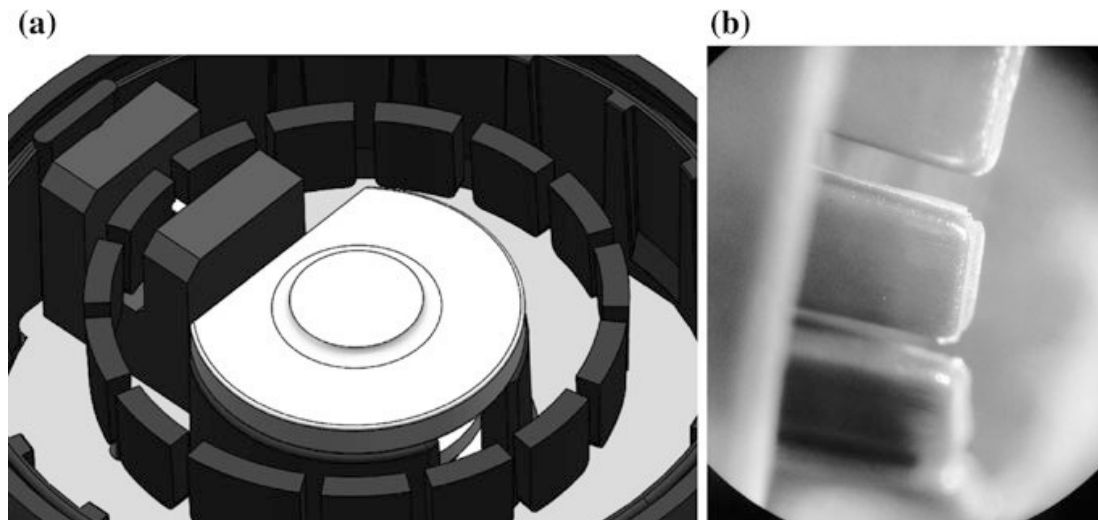


Fig. 10.5 Rotating speed and position detector. **a** Teeth and detector. **b** Detail of teeth

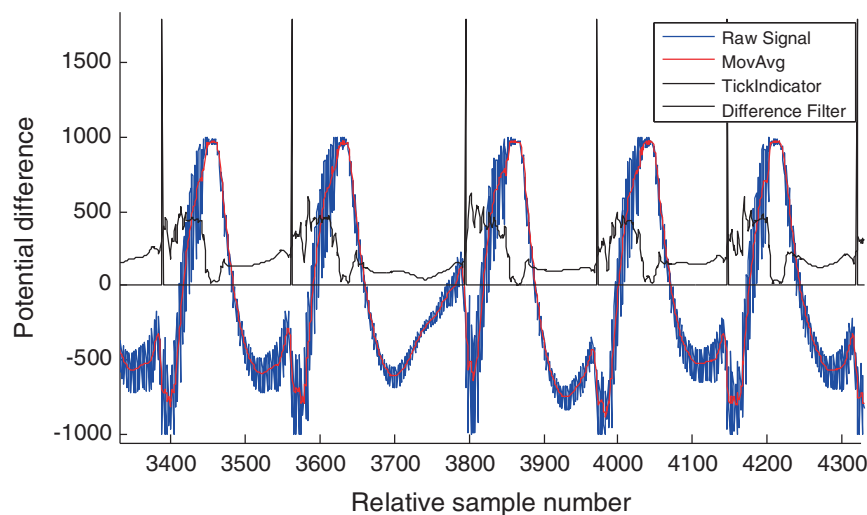


Fig. 10.6 Signal produced by the sensor of Fig. 10.5

differences between the peaks and the troughs on the signal in order to calculate when a tooth has passed the detector. However, at certain geometric variations and at higher velocities the detector can fail to identify the gaps between the teeth.

The above is quite a straightforward case where it is possible to do some simple robustness optimization so that the correct teeth numbers and dimensions are designed so that the detector works over a wide range of velocities. The robust optimization can also be undertaken for the microvariation of tooth dimension and detector positioning and even the influence of noise factors such as temperature causing thermal expansion. However, it is much easier for blind spots to occur between domains where it is unlikely that the mechanical engineer will fully appreciate the consequence of geometric variation on the electrical domain, and

the electrical engineer will often be unaware of the potential sources of mechanical variation and the decisions that induce it.

This lack of understand between the domains is significantly compounded when the complexity scales up. Design changes and optimizations can occur in each discipline with not full appreciation of the effect on the other. A very good example can be seen in the following case.

10.3.5 *Modelling Across Disciplines*

While there have been many valuable attempts to model mechatronics from a number of different authors, the design and modelling of mechatronics is still a considerable challenge [18]. In product development, an incorrect product integration within a system is clearly seen in the media. For instance, in 2014 the GM ignition switch (see Fig. 10.7) recall was headline news, with record breaking costs/damages to the company of around \$1.2 billion involving the recall of 28 million vehicles. The basic failure mode meant that the switch was unable to provide the torque to hold the key in the ON position while the vehicle was running, and in some circumstances the switch would slip from the ON to the (ACC)ESSORY position.

The failure to identify and remedy the issues in this case were quite systemic and reported at length by Valukas [19]. First, it is important to point out that this was a component provided to GM by a supplier and which had to fit on multiple GM vehicles. Both the switch and the series of vehicles each have their own electromechanical systems which must work in harmony. In the case of the GM ignition switch the mechatronic modelling approaches simply did not support the engineers adequately, leading to major consequences. The following three pieces of evidence are taken from the Valukas report to support this argument:

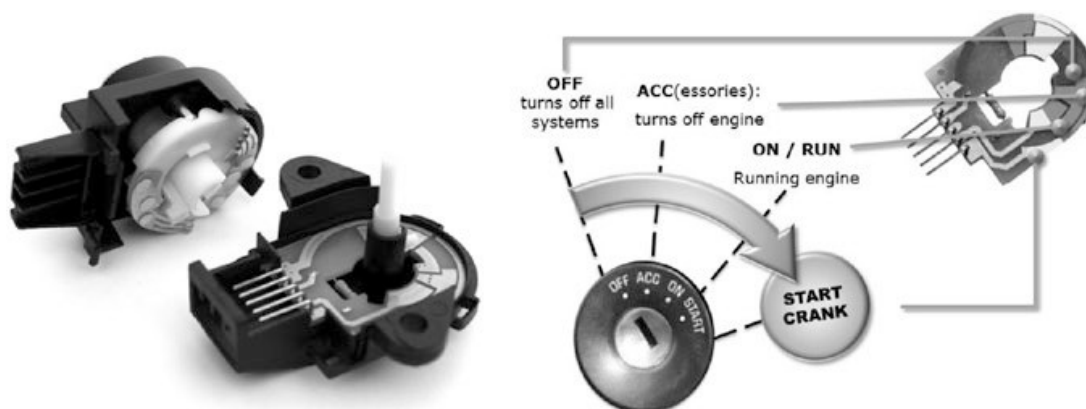


Fig. 10.7 Recalled GM ignition switch

1. None of the mechanical engineers working on the ignition switch were aware that moving the switch from the ON to the ACC position would shut off the airbag along with the power-assisted brakes and steering.
2. Due to electromechanical issues, the switch was sometimes unable to be operated in cold weather. This was then confused with the slipping issue.
3. Delphi had not achieved the required torque for the ignition switch. Given the switch's history of electrical failures, however, they were hesitant to make any changes that might jeopardize the functionality of the switch's architecture.

The first item mentioned in the report is about understanding the functionality of the products across domains. Without proper modelling of the functionality, engineers face the challenge of being able to foresee problems caused by decisions taken in one domain affecting other domains. This issue is not only present for engineers working on systems with many components such as cars, but is also present in products with fewer components and less complexity. The reason being that most companies are driving performance of their products in the pursuit of being one step ahead of their competitors. Inherently, this will result in many issues to be solved within each of the domains. Focusing on the problems lying ahead within each domain tends to attract engineers' attention to their own discipline-specific issues and away from integration issues; the so-called '*silo thinking*'. A failed interface between two domains may, however, just as well lead to a break down or degraded performance of a product.

The robustness of a mechatronic product can be affected in one of the three following ways:

1. Robustness issues within a single domain. For instance, feedback issues in electronic circuits creating unstable performance.
2. Robustness issues in the functional interaction between domains. For example, a mechanical strain gate creates an out of specification signal at one end of the performance spectrum, which creates an incorrect signal error when processed by the electronics.
3. Robustness issues caused by adverse effects. Adverse effects are unforeseen events creating an effect in domain other than where the event appears. Thus, heat from electronic components could cause a bearing to lock unintentionally due to the thermal expansion of materials.

To increase robustness, each domain must be internally robust (intra-domain robustness) as well as the product have interfacing functions between domains which are equally robust (inter-domain robustness). Some products may have a clear-cut interface between domains posing the opportune situation that the development can be divided into two different tracks with very little interaction (e.g. between mechanical and electrical development teams). The trend, however, seems to go in the opposite direction; namely to create more integrated products with many functional interfaces between mechanical, electronics and software solutions. This seems to have been the trend over the last decades and one might assume that this trend will continue in the future.

So to be able to create robust products in the future we need methods to access the inter-domain robustness as it seems that intra-domain robustness is better understood, modelled and comprehended, although not perfected. Tools to assess the level of complexity of the functional interfaces between domains as well as evaluating the vulnerability of a single functional interface and its effect on the robustness of the overall system are lacking. Fault tree analysis is one option but is far from adequate in assessing the robustness of a full-blown mechatronic system.

This speaks to the argument that the future for mechatronics research will respond to the modelling requirement to identify failure modes and the effects of variation across disciplines. A comprehensive modelling approach would lead to more effective design change management and design optimization.

10.3.6 The Importance of a Cybernetic Perspective on Robust Design?

In summary, it is quite clear that robustness optimization processes within a single domain, not only give an incomplete picture when optimizing a product, may actually reduce the cross-domain robustness for cybernetic products. This is not a minor consideration by any means, as the cross-domain robustness issues are seen as many both critical and main functions in many modern products. Examples of which are drone technology, self-driving cars, production robots or the exploding number of cybernetic products classed within the Internet of Things, all of which are characterized for a need of mechanical precision, timing and performance based on sensor inputs, signal processing. It would seem that defining cross-domain modelling approaches that can be used in conjunction with new and existing robust design approaches, not only defines the forefront of robust design research but also the quality and reliability of many emerging CPS.

10.4 Challenges for an Integrated Design Methodology for CPS

Design of CPSs involves close examination and further development of design methods, design processes, models and tools. The current trend in mechatronics involves networked mechatronic systems, or cyber-physical systems (CPS). Therefore product lifecycle management and product data exchange play an important role in CPS design. In order to push the performance of CPS and the related design process, it is necessary to increase the research on system modelling, thus significantly improving the system view. There are many challenges for future research towards improving the efficiency and quality of CPS product development. In terms of model-based methods and tools in early design phases

which are the themes of a Design Society Special Interest group [20] who emphasize challenges related to the following topics:

- Methods to generate solution concepts for CPS.
- Methods for evaluation of solution concepts.
- Methods for early concept optimization.
- CPS modelling aspects in terms of model integration, modular modelling, and model interfaces.
- Performance of CPSs (multivariable key performance indicators, methods for cross-domain aggregation for performance evaluation).
- Methods and tools supporting CPS Design.
- MBSE for CPS.
- Integration of the cyber and the physical sub-systems.
- Cyber-physical robustness.
- Information and knowledge flow during CPS design processes.
- Modelling engineering change propagation and functional couplings in CPSs.
- Description and modelling languages for all design phases of CPS.
- Closing the gap between scientific approaches and industrial practice.

The key to an integrated design methodology for CPS is modelling and simulation (see Fig. 10.8). In this context design models and their interchangeability between different design tools are very important during the design process. From the mechatronic design process viewpoint, models are containers of the knowledge of the product during its total life cycle. Simulations are producing information of the design problem. This may improve product knowledge and potentially also the quality of many analyses and decisions. The presented approach relies on modular model architecture and enables innovative design, flexibility, speed and assistance in nonroutine design questions.

The main point of view of the work is “*simulation*”, namely modelling and virtual experimentation regarding the behaviour of a system. The considered objectives of the simulation are:

- The creation of executable models of design concepts at the system level and the evaluation of alternative concepts.
- The creation and production of multidisciplinary simulation models at the system level of CPS.
- Simulation models for the use in system testing.

Linked to these objectives is the question of the required (mathematical) description of different types of simulation models. In this context, the aspect of usage and reuse of (simulation) models is an important topic. There is a lack of methods and software tools supporting the modelling and simulation aspects in the early phases of product development, during which detailed models due to incomplete information cannot be established and therefore the system has to be modelled on a level with high abstraction. It is a challenge for the future to derive the requirements for such tools and to develop appropriate software for this purpose.

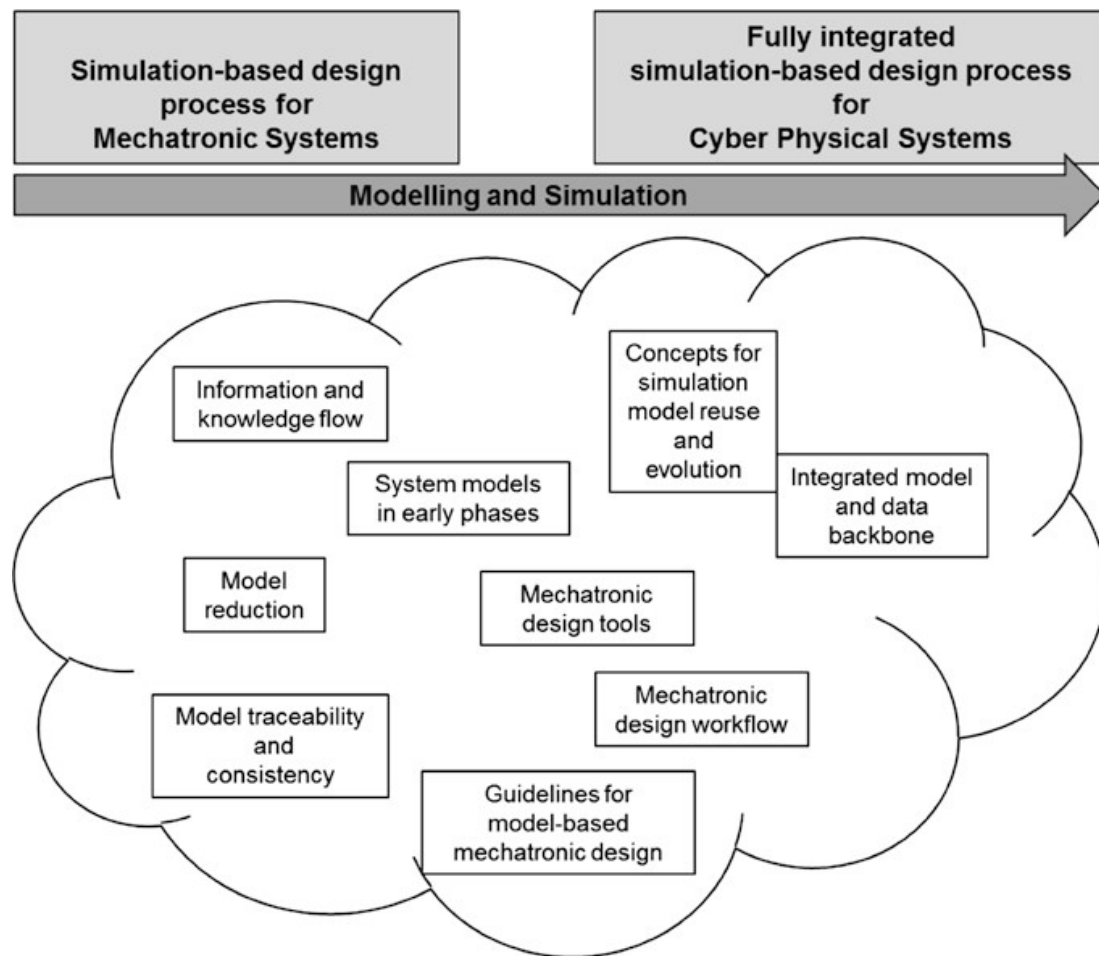


Fig. 10.8 Challenges for the future

For a simulation-based engineering approach, a model-based description of the system, with respect to its sub-systems, under consideration is a prerequisite. Especially for mechatronic systems this addresses:

- The models of the sub-systems.
- The integration of these models to an overall CPS—system model.

This engineering approach facilitates a holistic view on the overall system and should be continuously applied even from the very beginning of product development. Thus it becomes possible to model aspects such as requirements, functions, behaviour and structure in an integrated way which is essential especially during the early phases of design.

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